

A COUNTEREXAMPLE TO CARTWRIGHT–CHAN’S PRINCIPAL 6×6 CRITERION FOR TREE RANK

ABSTRACT. Cartwright and Chan ask, in the sixth of the seven open questions of [1], whether a matrix has tree rank 2 if and only if all of its principal 6×6 submatrices have tree rank at most 2. We refute the “if” half. The witness is the 7×7 zero-one dissimilarity matrix M whose one-entries are exactly the six pairs 13, 23, 45, 46, 57, 67: its tree rank is 3, while each of its seven principal 6×6 submatrices has tree rank exactly 2. All tree ranks are computed from the source paper’s own criterion for zero-one matrices.

1. THE QUESTION

Cartwright and Chan [1] study three notions of rank for symmetric matrices. A *dissimilarity matrix* of order n is a map from the 2-element subsets of $[n] = \{1, \dots, n\}$ to $\mathbb{R}$, written as a symmetric matrix with no diagonal entries. A *tree matrix* is a point of the tropical Grassmannian $G(2, n)$ of [3], i.e. a dissimilarity matrix satisfying the three-term Plücker relations. The *tree rank* $\text{tree-rank}(M)$ of a dissimilarity matrix M is the least k such that M is the entrywise minimum of k tree matrices. The last section of [1] lists seven open questions; the sixth reads, verbatim:

Does a matrix have tree rank 2 if and only if all its principal 6×6 submatrices have tree rank at most 2? Pachter and Sturmfels have made the same conjecture for their definition of tree rank [2, p. 124].

Both sides are read as “at most 2”; with “exactly 2” on the left the statement is refuted by any tree matrix, a degeneracy we do not use. Nothing in the sentence bounds the order, so it quantifies over all n and all real dissimilarity matrices of order n .

The “only if” half is a theorem and is not what falls. The three-term Plücker relations involve four indices at a time, so a principal submatrix of a tree matrix is a tree matrix, and the entrywise minimum commutes with restriction; hence tree-rank never increases when a principal submatrix is taken. The open half is the local-to-global “if”, and it is false. The Pachter–Sturmfels analogue named in the same sentence is for a different notion of mixtures, and nothing below touches it.

2020 *Mathematics Subject Classification*. 14T15, 15A80, 05C70.

Key words and phrases. tree rank, tropical Grassmannian, dissimilarity matrix, complete multipartite cover, counterexample.

2. THE WITNESS

Throughout, for a 0/1 dissimilarity matrix M of order n we write G_M for its *zero-entry graph* on $[n]$: $\{i, j\}$ is an edge if and only if $M_{ij} = 0$. Given pairwise disjoint (not necessarily covering) sets $I_1, \dots, I_k \subseteq [n]$ with $k \geq 2$, the associated *complete multipartite* graph has an edge between every two elements of distinct I_a ; vertices in no part are isolated. Equivalently, a graph is complete multipartite when, *among the vertices incident to some edge*, non-adjacency is a transitive relation. Write $r(G)$ for the least number of complete multipartite subgraphs of G whose union contains every *edge* of G (vertices need not be covered).

Every tree rank below is computed from the following proposition of [1] on 0/1 matrices, which we do not reprove. The statement is our wording, not a quotation, and it fixes the two conventions just given: parts need not cover $[n]$, and a cover is a cover of edges. The values 3 and 2 obtained below are the values this proposition assigns under those conventions.

Proposition 1 ([1], proposition on 0/1 matrices). *Let M be a 0/1 dissimilarity matrix of order n and let $r = r(G_M)$. If G_M has at most one isolated vertex then $\text{tree-rank}(M) = r$; otherwise $\text{tree-rank}(M) = r + 1$.*

Theorem 2. *Let M be the 0/1 dissimilarity matrix of order 7*

$$M = \begin{pmatrix} * & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & * & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & * & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & * & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & * & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & * & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & * \end{pmatrix},$$

that is, the matrix whose one-entries are exactly the six pairs 13, 23, 45, 46, 57, 67. Then $\text{tree-rank}(M) = 3$, while each of the seven principal 6×6 submatrices of M has tree-rank exactly 2. Consequently the “if” half of the sentence quoted in §1 is false, and the biconditional is refuted.

The zero-entry graph $G = G_M$ has the fifteen edges

$$12, 14, 15, 16, 17, 24, 25, 26, 27, 34, 35, 36, 37, 47, 56,$$

the remaining six pairs 13, 23, 45, 46, 57, 67 being the non-edges; fifteen and six exhaust the $\binom{7}{2} = 21$ pairs. With $A = \{1, 2, 3\}$ and $B = \{4, 5, 6, 7\}$, G is the complete bipartite graph between A and B — all twelve cross pairs — together with the single edge 12 inside A and the two disjoint edges 47 and 56 inside B . Its degree sequence is $(5, 5, 4, 4, 4, 4, 4)$, so $\delta(G) = 4$; in particular neither G nor any $G - i$ has an isolated vertex, and Proposition 1 reads $\text{tree-rank} = r$ throughout the proof.

Proof that $r(G) \leq 3$. Three complete multipartite subgraphs of G cover $E(G)$:

member	parts	#edges
c_a	$\{1, 2, 3\}, \{4, 5, 6, 7\} \rightarrow 14, 15, 16, 17, 24, 25, 26, 27, 34, 35, 36, 37$	12
c_b	$\{1\}, \{2\}, \{4\}, \{7\} \rightarrow 12, 14, 17, 24, 27, 47$	6
c_c	$\{5\}, \{6\} \rightarrow 56$	1

The three members contribute $12 + 6 + 1 = 19$ edge slots, of which four are duplicated — c_b repeats 14, 17, 24, 27 from c_a — so the union has $19 - 4 = 15$ edges and is exactly $E(G)$. $\square$

Proof that $r(G) \geq 3$. Suppose $E(G) = c_1 \cup c_2$ with c_1, c_2 complete multipartite subgraphs of G ; taking $c_2 = c_1$ is allowed, so the case $r(G) \leq 1$ is included in what follows.

(i) *No complete multipartite subgraph c of G contains both 47 and 56.* If it did, then 4, 7 lie in distinct parts of c and so do 5, 6. Since 45 is a non-edge of G it is a non-edge of c , so 4 and 5 lie in one part; since 46 is a non-edge, 4 and 6 lie in one part. Hence 5 and 6 lie in one part, contradicting $56 \in c$. This already excludes $r(G) \leq 1$. Relabelling, we may take $47 \in c_1$ and $56 \in c_2$.

(ii) *The member containing 12 does not meet vertex 3.* Say $12 \in c_i$. As 13 and 23 are non-edges of G , they are non-edges of c_i ; in the triple (1, 3, 2) the outer pair is an edge of c_i while both inner pairs are not, so if 3 were incident to an edge of c_i , non-adjacency would fail to be transitive among the vertices of c_i incident to an edge, and c_i would not be complete multipartite. Hence 3 is isolated in c_i and all four of 34, 35, 36, 37 lie in the other member.

(iii) *The case $12 \in c_1$.* Then $34, 35, 36, 37 \in c_2$, which also contains 56. Inside c_2 vertex 3 is adjacent to each of 4, 5, 6, 7, so those four lie in parts other than that of 3. Now 57 is a non-edge of G , so 5, 7 lie in one part of c_2 , and 67 is a non-edge, so 6, 7 lie in one part. Hence 5, 6 lie in one part, contradicting $56 \in c_2$.

(iv) *The case $12 \in c_2$.* Then $34, 35, 36, 37 \in c_1$, which also contains 47. As in (iii), 4, 5, 6, 7 lie in parts other than that of 3; 45 a non-edge puts 4, 5 in one part, and 57 a non-edge puts 5, 7 in one part, so 4, 7 lie in one part, contradicting $47 \in c_1$.

Both cases are impossible, so $r(G) \geq 3$. With the upper bound, $r(G) = 3$ and, G having no isolated vertex, $\text{tree-rank}(M) = 3$. $\square$

Proof that every principal 6×6 submatrix has tree-rank exactly 2.

Deleting index i leaves the zero-entry graph $G - i$, with $|E(G - 1)| = |E(G - 2)| = 10$ and $|E(G - i)| = 11$ for $i = 3, \dots, 7$. For each i the following two complete multipartite subgraphs of $G - i$ have union exactly $E(G - i)$; the parts are listed first, and after the arrow the edges they

contribute.

i	first member	second member
1	$\{4\}, \{7\}, \{2, 3\} \rightarrow 47, 24, 34, 27, 37$	$\{5\}, \{6\}, \{2, 3\} \rightarrow 56, 25, 35, 26, 36$
2	$\{4\}, \{7\}, \{1, 3\} \rightarrow 47, 14, 34, 17, 37$	$\{5\}, \{6\}, \{1, 3\} \rightarrow 56, 15, 35, 16, 36$
3	$\{4\}, \{7\}, \{1, 2\} \rightarrow 47, 14, 24, 17, 27$	$\{1\}, \{2\}, \{5\}, \{6\} \rightarrow 12, 15, 16, 25, 26, 56$
4	$\{1, 2, 3\}, \{5, 6, 7\} \rightarrow 15, 16, 17, 25, 26, 27, 35, 36, 37$	$\{1\}, \{2\}, \{5\}, \{6\} \rightarrow 12, 15, 16, 25, 26, 56$
5	$\{1, 2, 3\}, \{4, 6, 7\} \rightarrow 14, 16, 17, 24, 26, 27, 34, 36, 37$	$\{1\}, \{2\}, \{4\}, \{7\} \rightarrow 12, 14, 17, 24, 27, 47$
6	$\{1, 2, 3\}, \{4, 5, 7\} \rightarrow 14, 15, 17, 24, 25, 27, 34, 35, 37$	$\{1\}, \{2\}, \{4\}, \{7\} \rightarrow 12, 14, 17, 24, 27, 47$
7	$\{1, 2, 3\}, \{4, 5, 6\} \rightarrow 14, 15, 16, 24, 25, 26, 34, 35, 36$	$\{1\}, \{2\}, \{5\}, \{6\} \rightarrow 12, 15, 16, 25, 26, 56$

A reader checks each row by confirming that every listed pair is an edge of G and that the two lists together are exactly the edge list of $G - i$. So $r(G - i) \leq 2$.

It is not 1, i.e. no $G - i$ is itself complete multipartite. For $i = 1, 2, 3$ the graph $G - i$ still contains both 47 and 56, as well as the non-edges 45 and 46, so step (i) of the previous proof — which used nothing but those four pairs — applies with $c = G - i$. For $i = 4, 5, 6, 7$ the graph $G - i$ contains the edge 12 and the non-edges 13, 23, and vertex 3 is still incident to an edge there (35, 36, 37; 34, 36, 37; 34, 35, 37; 34, 35, 36 respectively), so step (ii) applies with $c = G - i$ and is contradicted. Hence $r(G - i) = 2$, and as $G - i$ has no isolated vertex, the corresponding principal 6×6 submatrix has tree-rank exactly 2. Together with $\text{tree-rank}(M) = 3$ this proves Theorem 2. $\square$

3. VERIFICATION

The program `verify.py` accompanying this note re-derives, in exact integer arithmetic and using nothing outside the Python standard library, the claims of §2 from the matrix as printed in Theorem 2: it reads M off the display, recomputes G and its non-edges, checks the three-element cover and the seven two-element covers member by member, and re-proves $r(G) \geq 3$ by an exhaustive search over all 209 nonempty complete multipartite subgraphs of G , so that the lower bound does not rest on the hand argument alone. The recorded run is `verify.output.txt`; it also contains checks of further facts about the 0/1 cell that this note does not claim. The refutation itself needs none of it: the argument of §2 is a page of finite combinatorics on the printed object.

Proposition 1 is taken as given, and no computation checks a transcription: the sentence quoted in §1 rests on the e-print cited below.

REFERENCES

- [1] D. Cartwright and M. Chan, *Three notions of tropical rank for symmetric matrices*, *Combinatorica* **32** (2012), no. 1, 55–84. doi:10.1007/s00493-012-2701-4. The open question and the proposition on 0/1 matrices are taken from the e-print arXiv:0912.1411v1, the only full text we could reach; we do not assert the wording of the printed text.
- [2] L. Pachter and B. Sturmfels (eds.), *Algebraic Statistics for Computational Biology*, Cambridge University Press, 2005; the analogous conjecture for their notion of tree rank is on p. 124. Not consulted here.
- [3] D. Speyer and B. Sturmfels, *The tropical Grassmannian*, *Adv. Geom.* **4** (2004), no. 3, 389–411. doi:10.1515/adv.2004.023.